

Nature research Code and Software submission information.

A. Compiled standalone software and/or source code

Available at:

https://github.com/szczotlab/nociceptive_pathways_revealed_by_activity_labeling/tree/main/Matlab/SC_GeneBrowser4.3DeploymentLite

B. Full dataset to load into GUI available at:

<https://zenodo.org/records/17392899>

1. System requirements: Software dependencies and operating systems:

Matlab R2024b (untested on earlier versions)

dependencies:

Image Processing Toolbox

Statistics and Machine Learning Toolbox

OS/ Microsoft Windows 11

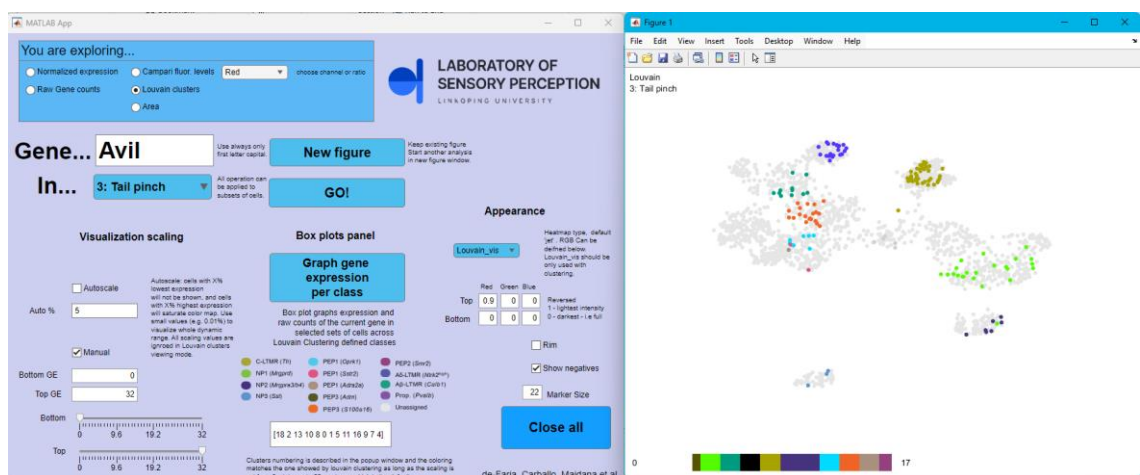
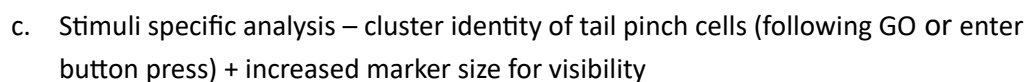
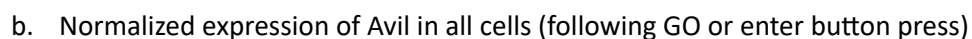
No non-standard hardware required

2. Installation guide available in dedicated QuickStart guide including instruction to run software with available data:

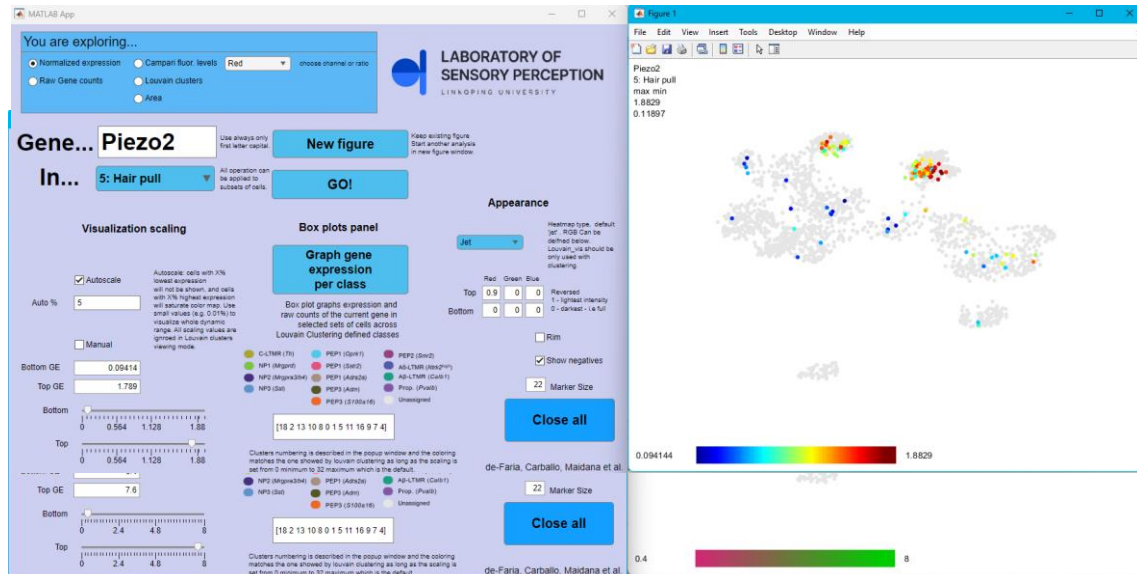
https://github.com/szczotlab/nociceptive_pathways_revealed_by_activity_labeling/blob/main/Matlab/QuickStart_SC_GeneBrowser.pdf

Typical install time under 5 minutes.

- a. Overview Louvain clustering (following GO or enter button press)

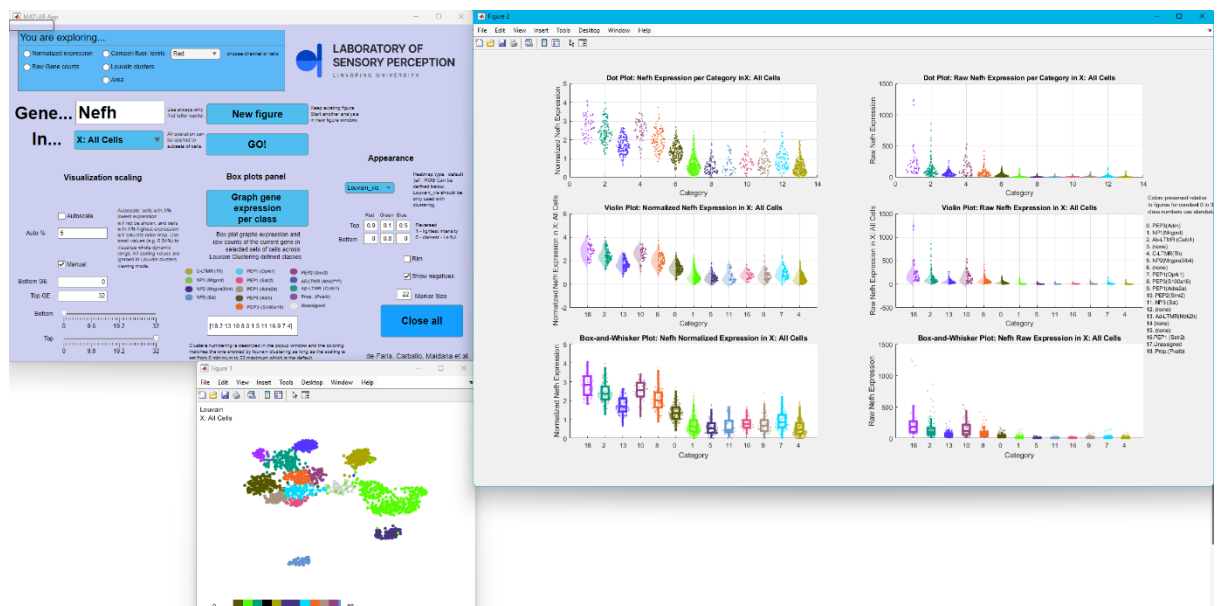


- d. Stimuli specific analysis – Normalized Piezo2 expression in of tail pinch cells (following GO or enter button press) + increased marker size for visibility



- e. Stimuli specific analysis – Raw UMI counts for Bmpr1b in control cells (following GO or enter button press) + custom RGB heatmap for scale

- f. Cluster wise gene expression analysis for Nefh (following 'Graph gene expression per class press'), note at least one UMap analysis window needs to be open before pressing the button.



- g. Cluster wise gene expression analysis for Uts2b in bladder distension responding cell (following 'Graph gene expression per class press'), note at least one UMap analysis window needs to be open before pressing the button.

